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Invited Review in Celebration of the 50th Anniversary of EURO

50 years of warehousing research—An operations research perspective

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ABSTRACT

Warehouses have always been an essential part of supply chains, but despite their fundamental role they were not seen as especially mission critical. With the advent of e-commerce, same-day deliveries, omni-channel retailing, and global supply chain disruptions, however, this assessment has changed, and today's warehouses have evolved to technology-enriched fulfillment factories with strategic relevance. This paper traces the evolution of picker-to-parts and parts-to-picker warehouses from basic systems of the first generation, via extended system setups of the second generation, up to state-of-the-art robotized distribution centers of the third generation. Specifically, we highlight the most influential scientific contributions of the operations research (OR) community within each generation that have supported this evolution over the past 50 years. Beyond the historical perspective, we outline an agenda for the warehousing research of the future.

1. Introduction

To decouple the time of production from the time of consumption, storage of goods in warehouses has always played a fundamental role in the history of mankind. References to warehouses (granaries) can, for instance, already be traced back to the holy books of the three major European religions, the Qur'an, the Torah, and the Bible. Since then, despite of the fundamental role of warehouses in almost any supply chain, warehouses and their operations have not always been assessed as especially mission critical. During the advent of phenomena such as e-commerce, same-day deliveries, and omni-channel retailing, but also as the result of recent supply chain interruptions, however, warehouses (and research on them) have gained momentum. Today's state-of-the-art distribution centers have evolved to technology-enriched fulfillment factories, which play a critical role to satisfy demanding customers with time-critical, cost-efficient, and reliable deliveries (see Boysen et al., 2019b).

To support our claim regarding the critical role of today's warehouses, to showcase the huge variety of different system setups, and to exemplify the diversity of managerial and research challenges coming along with them, we start our elaboration with a sketch of four example e-commerce warehouses, schematically depicted in Fig. 1.

To enable within-the-hour deliveries of urban customers with convenience goods, special facilities located directly in the city centers are applied by many retailers such as Carrefour (France), Mercadona (Spain), and Rappi (Brasil) (Interlake Mecalux, 2021), as well as quick-commerce providers like Getir and Flink. Since these facilities only service online orders and are not accessible for walk-in customers, they are also denoted as *dark stores*. Land in metropolitan areas is a *scarce* and costly resource, so that the typical dark store is restricted to a small footprint. Hence, it can offer only a reduced assortment of frequently-ordered stock keeping units (SKUs) (Seidel, 2021). To compensate for the high delivery costs on the last mile, many e-grocers demand a minimum order value or charge an extra delivery fee, so that the typical order sizes in this segment exceed a dozen of SKUs (Seidel, 2021). Two different types of warehousing systems seem especially suitable for this purpose:

(a) Dark stores can mimic the layout of conventional stores, where SKUs are stored in person-high shelves, accessed by human pickers. They follow the picker-to-parts paradigm and walk to the storage locations of ordered SKUs. A picker is typically equipped with a manual picking cart that carries multiple bins, each dedicated to a customer order, that are processed in parallel on the same picking tour (i.e., according to the sort-while-pick policy, see Gademann & Van de Velde, 2005). Dark stores operate under significant time pressure, and urgent orders pop up dynamically over time. In this context, solving the resulting order batching problem (i.e., which orders should be processed during the same picking tours, see Section 3.2) in a highly dynamic environment, where the picking process must furthermore be coordinated with the tours of last-mile delivery, is a challenging ongoing research task (e.g., see Kuhn et al., 2021).

(b) To use scarce urban land more efficiently, compact storage systems, such as those offered by Autostore (see Azadeh et al., 2019a),

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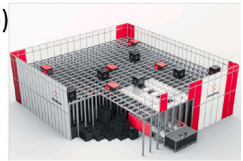
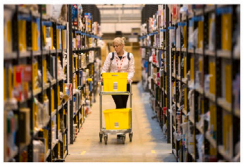
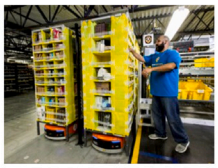
Purpose	Characteristics	Picker-to-parts	Parts-to-picker
Inner-city dark stores for groceries	small assortment, large orders, within-the-hour delivery	(a)	(b) 
Rural e-commerce distribution centers	huge assortment, small orders, next-day delivery	(c) 	(d) 

Fig. 1. Examples of e-commerce warehouses: (a) Dark store based on manual picking (Source: Stop & Shop), (b) Micro fulfillment based on Autostore (Source: Swisslog), (c) Manual picking in a scattered storage warehouse (Source: Geoff Robinson), (d) Picking workstation supplied by shelf-lifting mobile robots (Source: Amazon).

can be applied. In these systems, bins filled with SKUs are stored on top of each other in stacks supported by a grid-shaped frame, on which autonomous robots travel. Equipped with a lifting mechanism, each robot can only access the topmost bin of a stack, which must be relocated whenever a "buried" bin of a lower level is demanded. Retrieved bins containing a requested SKU are delivered to a lift that services a picking workstation, where a human picker obtains the demanded product and subsequently fills order bins. Since the stationary pickers are supplied with demanded SKUs, these systems follow the parts-to-picker paradigm. The coordination of multiple robots and orders that impact each other via relocated bins and blocked travel paths on top of the grid in a highly dynamic environment is another ongoing research challenge for the warehousing community (see Section 4.3).

The complete assortment of online retail, which due to the "long-tail" phenomenon (Brynjolfsson et al., 2003) is typically much larger than that of store-based retail (Boysen et al., 2021), is serviced from vast distribution centers located in remote areas with good transport connection. Rural locations reduce land and labor costs, but induce longer delivery times, so that B2C e-commerce customers have learned to expect next-day deliveries for their small-sized orders (e.g., an average order at Amazon Germany contains just 1.6 items, see Boysen et al., 2019b). The following two warehouse setups can frequently be observed for this purpose:

(c) Many e-commerce retailers extend traditional picker-to-parts warehouses (i.e., order-by-order picking from unit-load racks) by batching, zoning, and scattered storage (Schiffer et al., 2022). In a wave-wise manner, many pickers, operating in parallel in multiple zones of the warehouse and each equipped with a manual picking cart that carries multiple bins, collect partial orders that all together constitute the current wave of customer orders. The demanded products are retrieved from scattered storage shelves where different individual products are widely mixed. The latter increases the probability that products demanded by the same picker on the same tour can be obtained without excessive unproductive walking. Once the bins of products are completed and handed over to a conveyor systems, they are transported toward a consolidation area where the wave is sorted according to customer orders and finally shipped. Other than the stacking of unit loads (e.g., pallets) into racks in a traditional warehouse, the replenishment process of individual products in scattered storage warehouses constitutes a laborious task that offers plenty of new ongoing research challenges (see Section 3.4).

(d) To avoid the unproductive picker walking of the previous setup, online retailers such as Amazon (Weidinger et al., 2018) or JD.com (Qin et al., 2022) also apply alternative warehousing systems, following the parts-to-picker paradigm. In these robotic mobile fulfillment systems, a large fleet of autonomous mobile shelf-lifting robots retrieve person-high inventory pods filled with a mix of different SKUs from a vast rack storage area and deliver them toward picking workstations where stationary pickers withdraw requested products. In this setup, a large number of products, pods, and picking workstations need to be coordinated in a highly dynamic environment (see Boysen et al., 2019b). This setup constitutes another ongoing challenge for the warehouse research community (see Section 4.3).

These examples suggest the following findings: (i) Even for the same purpose (similar assortments, order profiles, and delivery lead times) very different warehousing systems exist. Often it is a picker-to-parts system competing with a parts-to-picker solution. These two alternatives heavily vary in their invest-to-operational-costs ratios. Hence, selecting the right combination of warehousing systems constitutes an important managerial task that demands scientific support. (ii) Warehousing systems for different purposes vary immensely in their setup, which is mainly influenced by the characteristics listed in Fig. 1. Hence, retailers having to service different types of products and customers have to decide for the right mix of warehousing systems they install within each single facility and how to spread them within their distribution networks. (iii) The arsenal of suitable research methods for supporting the wide spectrum of warehousing decisions, ranging from the strategic choice of the right warehousing systems via tactical decisions addressing the right storage policy up to operational decisions regarding the scheduling of pickers and mobile robots, is also vast.

It is the purpose of this survey paper to structure the plethora of systems, decisions, and research methods that are relevant in the warehousing domain. To do so, we take the historical perspective and trace the evolution of warehousing over time. Specifically, we differentiate the two basic types of warehousing structures, namely picker-to-parts and parts-to-picker systems, and trace their evolution by distinguishing different generations. Starting with the basic system setups of the first generation, we then elaborate the extended system setups of the second generation, and end with the state-of-the-art robotized distribution centers of the third generation. We describe the differentiation of these generations in more detail in Section 2, where we define the scope of our survey. Then, Sections 3 and 4 treat the three generations of picker-to-parts and parts-to-picker warehousing systems, respectively. For both systems types and each generation, we define the system setup, describe the main decision problems, elaborate seminal research contributions, and sketch future research needs. Finally, Section 5 concludes the paper by outlining a research agenda that goes beyond our generation-based historical perspective.

2. Scope of review

Basic definitions: Warehousing is defined as the intermediate storage of goods between stages in a supply chain Bartholdi and Hackman (2019) and consists of the basic functions receiving, storage, order

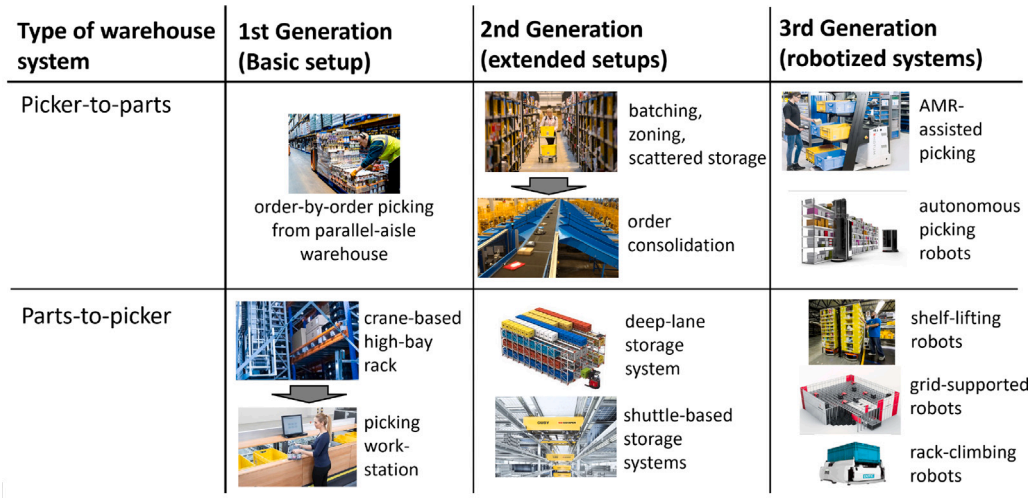


Fig. 2. Framework for our survey: The three generations of picker-to-parts and parts-to-picker warehousing systems.

picking, and shipping (Gu et al., 2007). Its main purpose, i.e., decoupling the time of production from the time of demand, enables various further company missions, such as realizing economies of scale in purchasing (e.g., quantity discounts) transportation (e.g., reduced travel), and production (e.g., larger lots) (see Lambert et al., 1998). Thus, warehousing is not independent of other supply chain stages. A *warehousing system* provides the environment to execute the basic warehousing functions listed above. It consists of hardware, which can be further subdivided into storage systems (e.g., a rack), material handling systems (e.g., a conveyor belt), and picking tools (e.g., a pick-by-voice solution), and processes defining the work flow along the applied hardware elements (e.g., sequential zone picking) (Boysen et al., 2019b). According to the principle how SKUs and pickers are brought together, warehousing systems are subdivided into *picker-to-parts* and *parts-to-picker* systems. In the former, the pickers must move toward the storage locations of demanded SKUs, and in the latter it is the SKUs that are brought toward stationary pickers by a material handling system. The (largely) human workforce of picker-to-parts systems makes them highly flexible but causes high operational wage cost. The fixed (and thus less flexible) machinery of parts-to-picker systems, instead, requires a larger one-time invest, but can save on operational expenses, which is especially valuable in high-wage countries. To delve deeper into the advantages and disadvantages of both systems, as well as their recommended domains of utilization, we recommend consulting Boysen et al. (2017) as well as Section 5. Note that we reduce the basic classification of warehousing systems to this dichotomy, although there are other review papers that suggest alternative approaches. Van Den Berg (1999), for instance, adds the class of picker-less warehousing systems, if no humans are involved (see, De Koster et al., 2007, for a similar, yet more detailed classification). In this paper, we do not follow these suggestions and still speak of a picker-to-parts system even if it is an automated picking device (e.g., an autonomous picking robot) that moves along the shelves to retrieve products. Analogously, we still speak of a parts-to-picker system even if the SKUs are transported toward a stationary automated picking device (e.g., an industrial palletizing robot). Further note that hybrid systems (e.g., where a robot fetches bins in a picker-to-parts process from shelves and delivers them to a workstation, where the demanded products are picked from the bins) are specifically discussed in Section 5.

Framework for structuring warehousing research: To structure the history of picker-to-parts and parts-to-picker warehousing systems, we differentiate three generations, schematically depicted in Fig. 2. Starting with the basic system setups of the 1st generation, we have extended system setups within the 2nd generation, and reach today's state-of-the-art warehousing systems based on robotized processes of the 3rd generation. Naturally, neither the erection of real-world warehouses nor research on them strictly follows our stylized generations over time. Even today, warehouses with a 1st generation setup are newly erected. Analogously, an exact algorithm with polynomial runtime for routing a single picker along a given set of storage locations in a parallel-aisle warehouse of the 1st generation with multiple cross aisles has only been introduced by Pansart et al. (2018). This is 35 years after this line of research was initiated by the seminal paper of Ratliff and Rosenthal (1983), but ten years after the 3rd generation of shelf-lifting robotized parts-to-picker systems of Fig. 1(d) have been introduced to the scientific literature by Guizzo (2008). Thus, neither the evolution of warehousing systems applied in real-world distribution centers nor the research on them strictly follows our generation differentiation on a timeline. Instead, there are leaps and time overlaps among generations. Nonetheless, we believe in the usefulness of these generations in order to structure the varied field of warehousing research.

For elaborating each system type and generation, we follow the same general structure in each section. First, we describe the system setup. Then, we define the basic decision problems and seminal research papers for each of the three main decision areas of warehousing (ordered according to planning horizon): (i) layout design planning, (ii) storage assignment, and (iii) order processing.

Modeling approaches: For these decision areas, different models are used. The commonly used models include travel time (TT) models, queueing network models, mathematical programming models, stochastic dynamic programming models, and simulation. Each model has its pros and cons, which makes them fit for specific decision types.

TT models are typically used for design-related decisions. They can be used to estimate the average amount of time it takes to execute a certain task (in a certain system using a specific policy). Since closed-form solutions can often be obtained, this makes it possible to optimize design choices. They are also often used as building blocks in queuing network models to estimate, e.g., service times. The limit of TT models is that they cannot easily deal with interactions between resources or queuing delays.

Queuing network models can deal with queuing delays. Different types can be distinguished: open (OQN), closed (CQN) and semi-open queuing networks (SOQN). In queuing networks, jobs (e.g., customer orders) undergo several tasks in a probabilistic sequence, with possible waiting at the different service stations. Different from an OQN, in a CQN the number of jobs remains constant: every time a job finishes, a new job enters. By their nature, these models particularly fit estimating throughput capacity for a given number of resources (e.g., robots, pallets, or carts), where these resources are reusable and circulate in the network. However, they cannot determine a job's throughput time, as

the waiting time before a job's release is not included. SOQNs combine the benefits of closed and open networks and are able to estimate both throughput capacity and job throughput time. Queuing models rapidly become intractable and the analysis is often only approximate.

Mathematical programming can deal with detailed operational decisions and can optimize them (e.g., by using a default solver) to achieve an objective. Typical objectives include minimizing the makespan of a given set of orders or the tardiness of jobs. Due to the combinatorial nature of many tactical and operational warehousing decisions (e.g., storage assignment, order batching, and picker routing) and the need to combine them due to interdependence, often heuristic methods must be applied to solve instances of real-world size.

Stochastic dynamic programming models are fit to take optimal operational decisions geared toward an objective when uncertainty is involved in a dynamic environment. However, these models suffer from the curse of dimensionality, which is why often approximate methods are needed for practical applications.

Simulation models can represent warehouse operations with all relevant decisions at a great level of detail. They are typically used to validate outcomes of the other decision models, as they cannot easily be used for system choice, system optimization, or policy evaluation; every system or policy requires a different model, and model development takes time.

In sum, all methods have their strengths, but also their disadvantages. Selecting the right modeling approach depends on the research question and the horizon of the decisions that must be taken.

Relation to previous surveys: Due to the increasing importance of warehousing it is anything but surprising that there already exist quite a few previous survey papers in this domain. For instance, there are the general surveys on the complete field by De Koster et al. (2007), Gu et al. (2007, 2010), Rouwenhorst et al. (2000), Van Den Berg (1999) and the following ones on specific warehousing topics: Azadeh et al. (2019a), Fragapane et al. (2021) (robotized warehouses), Boysen et al. (2019b) (e-commerce warehouses), Boysen et al. (2021) (warehouses for store-based retail), Agatz et al. (2008) (e-fulfillment in a multi-channel environment), Masae et al. (2020) (picker routing), Casella et al. (2023) (order picking trends),Gagliardi et al. (2012), Roodbergen and Vis (2009) (automated storage and retrieval systems), Boysen et al. (2023b) (synchronization problems in parts-to-picker systems), Boysen and Stephan (2016) (crane scheduling in storage systems), Van Gils et al. (2018) (holistic research combining multiple planning tasks), and Gong and De Koster (2011) (stochastic warehousing research). None of these review papers has taken the historical perspective, which is our main justification for yet another survey paper. Further arguments for our historical perspective are the following:

(i) Other than maybe in historical science, where it is an ongoing debate whether something can actually be learned from the past for future (e.g., political) decisions (e.g. Carr, 2003), we are convinced that warehousing research should be aware of its past. There are a lot of technical developments (e.g., new storage systems or robots) that slightly alter the system setup. Researchers, having to design suitable decision support tools, cannot always restart from scratch and should carefully observe what techniques that worked well in the past might be extendable with reduced effort. (ii) Although warehousing systems may vary significantly in their outer appearance, they may be accessible with similar models. For instance, a human picker workforce is certainly bound to much larger variation in picking times and walking paces than robotized picking devices. Hence, the former rather suggests stochastic models to better accommodate this variation (De Lombaert et al., 2023; Grosse et al., 2017). For standardized robots, instead, deterministic models should only lead to a rather negligible loss of precision. Despite of these undeniable differences, legislation and strong labor unions often veto the tracking of individual workers and stipulate the application of standard times instead. In this case, the same type of deterministic models for picker routing and batching can be applied for very different system setups. Knowing the past is a prerequisite for these transfers. (iii) Finally, EJOR has contributed significantly to the progress of warehousing research. For instance, 13 of the 18 above listed survey papers from the warehousing domain have been published in EJOR, and 286 EJOR papers use the terms "warehouse" or "warehousing" either in the title, the abstract, or the keywords. Thus, our historical view on warehousing is also an opportunity to honor the contributions of EJOR in the past 50 years to this field of research.

Literature retrieval: Finally, we briefly sketch our literature retrieval policy (or rather, we openly admit the lack of one). There is abundant OR literature on warehousing. Furthermore, there are already some excellent review papers, which aim at a comprehensive survey of (some specific aspect of) warehousing. Therefore, we have decided against a structured database search like it is, for instance, advocated by Hochrein and Glock (2012). Instead, we only select a few seminal papers of each generation that influenced warehousing research in a significant manner (e.g., by inspiring plenty of subsequent research or establishing a new research area) according to our own subjective perception.

3. Picker-to-parts systems

In picker-to-parts systems, it is the (either human or robotized) pickers who move toward the storage locations of the SKUs in order to retrieve the requested number of products defined on (either paper-based or electronic) pick lists. It seems hardly possible to precisely trace back the historical roots of picker-to-parts warehouses. However, since it is the rather natural setup how to structure warehousing processes, it seems likely that already ancient warehouses (e.g., those of ancient Rome, see Michael, 2016) followed this basic setup. Three representatives for the three generations of picker-to-parts warehouses, which we elaborate in the more detail in the following, are depicted in Fig. 3.

3.1. 1st generation

System setup: The main hardware elements that constitute the 1st generation of picker-to-parts warehouses are racks, typically for standardized unit loads such as pallets, that are arranged along both sides of parallel picking aisles. These picking aisles are connected by cross aisles, which are located at the front and back. Occasionally, there are additional inner cross aisles to shortcut between adjacent picking aisles. In what follows, we call this standard rack layout the *parallel-aisle structure* of warehouses. If multi-story racks are applied, like they are depicted in Fig. 3(left), they are typically divided into a picking level on the ground floor and an upper reserve area. To resupply the former from the latter, material handling devices such as forklifts or reach trucks are applied to replenish pallets. The human picker workforce either walks accompanied by a manual picking cart or drives on a motorized picking cart to the picking positions defined in the current pick list. In the basic case, which we assume for the 1st generation, pick lists are processed in an order-by-order manner. This means that each customer order constitutes a separate pick list and is thus collected on a separate picking tour. Each tour starts and ends at a single central depot where the orders are collected and prepared for shipping.

Layout design: The default layout of a traditional rack-based storage area is the parallel-aisle structure. There are, however, also alternative layouts, which violate what Gue and Meller (2009) call the *unspoken rules* of layout design: "1. Picking aisles must be straight, and parallel to one another. 2. If present, cross aisles must be straight, and they must meet picking aisles at right angles." The most prominent non-conventional layouts are called flying-V (Gue & Meller, 2009) (i.e., from a bird's eye view the inner cross aisles cut a 'V' through the parallel picking aisles with the lowest point of the 'V' being the depot), fishbone (Çelik & Süral, 2014) (i.e., the picking aisles below the 'V' are shifted by 90°), and discrete cross aisle design (Öztürkoğlu

Fig. 3. Generations of picker-to-parts warehouses: Parallel-aisle structure of 1st generation (Left; Source: Waredock), different zones of a scattered storage warehouse of the 2nd generation (Middle; Source: Álvaro Ibáñez), and autonomous picking robots of the 3rd generation (Right; Source: Magazino).

et al., 2012) (i.e., the inner cross aisles do not cut through the parallel picking aisles in a straight manner but at various positions). There is no clear hierarchy among the parallel-aisle structure and these non-conventional warehouse layouts, whose performance strongly vary with the pick list structure (Çelik & Süral, 2014). However, the vast majority of real-world warehouses applies the parallel-aisle structure.

Another layout choice (implicitly) contained in the parallel-aisle structure is the single depot that constitutes the unique start and end point of each pick tour. Especially, many modern e-commerce warehouses, however, apply a decentralized depot structure in their vast picking zones (Weidinger et al., 2019). Here, a depot is only one of many access points to the central conveyor system that transports filled bins toward a central consolidation and shipping area. Multiple depots increase the flexibility where to start picking tours, and the potential gains of such a decentralized depot structure have, for instance, been investigated by De Koster and Van Der Poort (1998), Weidinger et al. (2019), and Schiffer et al. (2022).

For 1st generation picker-to-parts warehouses, however, we presuppose the parallel-aisle structure with a single central depot. Once the choice for this basic layout is made, the main layout-related decision problem that remains is called the *aisle configuration problem* (see De Koster et al., 2007). Here, the number, length, and width of picking and cross aisles must be determined. The main trade-off of this problem is related to the right number of cross aisles: More cross aisles increase the flexibility to shortcut into adjacent picking aisles but supplant storage capacity. For a given storage capacity they, hence, increase the footprint of the storage area. Given a certain storage capacity that must fit into the storage area, the aisle configuration problem seeks decisions on the above layout-related issues in order to maximizes a throughput related objective. The latter, however, not only depends on the layout but also on the storage assignment, the picker routing, and the order structure that must be processed. To successfully consider this manifold interdependence, holistic research methods based on simulation (Petersen, 2002), data envelopment analysis (Chen et al., 2010), and mathematical programming (Ozden et al., 2021) have been applied.

Storage assignment: When deciding on the specific position in the storage area of a warehouse where each SKU is to be stored, there are basically six alternative storage assignment policies to be differentiated (see De Koster et al., 2007): (a) random storage, (b) closest-open-location storage, (c) dedicated storage, (d) full-turnover storage, (e) class based storage, and (f) family grouping.

(a) Under the *random storage* policy, for each incoming unit load its location in the warehouse is randomly selected among all currently available empty storage locations. This ensures a high space utilization at the expense of increased travel distances (e.g., see Petersen, 1997).

(b) To reduce the latter, *closest-open-location storage* leaves the choice of the storage location for an incoming unit load not to chance, but always assigns the currently empty storage location closest to the depot. This ensures densely packed shelves close to the depot, whereas open storage positions are moved to the back of the warehouse (Fukunari & Malmborg, 2008).

(c) If each SKU is stored at a fixed storage position, the warehouse follows the *dedicated storage* policy. On the positive side, fixed positions allow pickers to familiarize with the storage layout, and it is possible to store heavy SKUs closer to the depot. The latter ensures that heavy products end up at the bottom of pallets or roll cages, where they do not endanger fragile SKUs (De Koster et al., 2007). On the negative side, the storage density is typically low, because storage space must be reserved (theoretically up to each SKU's maximum stock level), even if these products are currently not at hand. Many pickers-to-parts warehouses of the 1st generation with racks divided into a floor-level picking area and an upper-level reserve area (see Fig. 3(left)) apply dedicated storage in the former and random storage in the latter area. This eases orientation for the pickers, while storage space in the reserve area is efficiently utilized.

(d) *Full-turnover storage* assigns SKUs with the highest sales rates the locations nearest to the depot, whereas slow movers are located toward the back of the warehouse. The most famous KPI (key performance indicator) to operationalize this general aim is the cube-per-order (COI) index introduced by Heskett (1963, 1964). The COI relates each SKU's total space requirement to the number of trips required to satisfy its demand per period. Problems with this approach occur with volatile product demands. They cause frequent effort to relocate products with altered priorities among storage locations (see Christofides & Colloff, 1973).

(e) Following the famous Pareto principle, according to which 80% of consequences come from 20% of "vital few" causes, *class-based storage* partitions the storage space into different classes ordered by increasing distance from the depot. The closest class is reserved for the vital few A-SKUs of fast movers, followed by less frequently ordered B-products, and so on. The products belonging to a class can, for instance, be derived via the COI index, and within each class random storage is often applied (De Koster et al., 2007). Different studies, evaluating the number of classes to be used and their exact arrangement within the aisles, are, for instance, provided by Petersen (1999, 2002) as well as Petersen and Aase (2004).

(f) If specific products are frequently ordered together, then it can be advantageous to store them closer together in order to reduce unproductive picker travel. A popular example are drills and boring heads that frequently end up in the same shopping baskets. Realizing this basic idea is intended by the *family grouping* policies. The methods to operationalize this aim reach from simple correlation-based rules (e.g., Frazelle & Sharp, 1989) via cluster-based approaches (e.g., Liu, 1999) up to exact optimization methods for given historical order sets (Boysen & Stephan, 2013).

Order processing: Among the most heavily cited figures in warehousing research are certainly the 55% of the total warehouse operating expense that can be attributed to order picking and the 50% share of the average picker's working hours used for traveling through the warehouse (De Koster et al., 2007; Tompkins et al., 2010). Although the empirical underpinning of these percentages is thin, still, reducing unproductive picker travel is one of the most important levers to cut costs (without deteriorating service) in picker-to-parts warehouses. The question how to efficiently route pickers in warehouses is, thus, one of the classics of warehousing research ever since (an in-depth survey on this topic is provided by Masae et al., 2020). For a given

pick list, the very basic picker routing problem equals the famous traveling salesman problem (TSP, e.g., see Applegate et al., 2011): Starting and ending each tour in the depot, the picker aims to visit all storage positions defined on the pick list on the shortest tour. For general graphs, the TSP is well-known to be strongly NP-hard (Garey & Johnson, 1979). The parallel-aisle structure of warehouses, however, induces specially structured distance matrices. This special structure is exploited by Ratliff and Rosenthal (1983), who provide an efficient routing algorithm that runs in polynomial time if there are only cross aisles at the front and back.

In a nutshell, the basic idea of their seminal algorithm is as follow. They first formulate the picker routing problem as a Steiner TSP, where some nodes must (i.e., the storage positions in the picking aisles) and some nodes can (i.e., nodes marking the meeting points of cross and picking aisles at the front and back) be visited. Then, Ratliff and Rosenthal (1983) show that in an optimal solution each picking aisle within such a graph can only be traversed in six different modes (i.e., (1) the aisle is not accessed at all, which is only possible if it contains no storage positions, the aisle is completely traversed (2) once or (3) twice, all storage positions of an aisle are visited either exclusively from (4) north or (5) south, or all storage positions of an aisle are partitioned among (6) two loops from north and south, where the turning point is at the largest gap between two neighboring positions). This reduced set of potential aisle traversals is then concatenated in an aisle-by-aisle manner by a dynamic program.

This algorithm has been extended by Roodbergen and De Koster (2001b) to also cover an additional inner middle cross aisle. Only recently, Pansart et al. (2018), based on the results of Cambazard and Catusse (2018) for rectilinear Steiner trees, have provided an efficient algorithm for an arbitrary number of additional inner cross aisles. Their algorithm has a runtime complexity of $\mathcal{O}(hn7^h)$, with n being the number of picking positions, which is in polynomial time if the number of cross aisles h is bounded by a constant. Further extensions of the algorithm of Ratliff and Rosenthal (1983) have been introduced by De Koster and Van Der Poort (1998) (decentralized handover of picked products to conveyors running along cross aisles), Masae et al. (2020b) (routing in a non-conventional layout), Masae et al. (2020a) (arbitrary start and end points), and Löffler et al. (2022) (clustered TSP with given cluster sequence).

If optimal solutions can efficiently be obtained, technically there is no need to accept non-optimal solutions of heuristics. There is, however, also a long-lasting tradition of research on heuristic picker routing methods. Prominent examples are, for instance, the S-shape (meander through the aisles in an S-shaped tour), return (never use the back cross aisle), mid-point (always return before crossing the middle of a picking aisle), and largest-gap (return in each picking aisle between those two positions having the largest gap among them) heuristics (for detailed descriptions, more heuristics, and performance tests, e.g., see Hall, 1993; Petersen, 1997; Roodbergen & De Koster, 2001a). It is often argued that the tours of these heuristics are more intuitive for human pickers, whereas optimal tours may "seem illogical or suboptimal to the order pickers who then, as a result, deviate from the specified routes" (Gademann & Van de Velde, 2005). It would be interesting to see whether the empirical findings summarized by Glock et al. (2017) for such a *maverick picking* still hold today, where more and more order pickers are digital natives used to algorithmic advice. To avoid maverick picking, many warehouses nowadays navigate pickers automatically via pick-by-voice solutions. Hence, routing heuristics have certainly lost some of their importance over the years, although physical constraints (e.g., picking carts that cannot easily turn in a narrow aisle) may still force pickers to deviate from theoretically optimal routes.

Another stream of research extends the basic picker routing problem by further aspects (for an exhaustive survey, see Masae et al., 2020). Among those aspects with the most practical relevance are certainly the consideration of precedence constraints among products (e.g., to prevent damage of fragile products, see Chabot et al., 2017), and blockings among multiple pickers in the same area of the warehouse (e.g. Schrotenboer et al., 2017).

At the end of this section, we would like to note that despite their long tradition 1st generation picker-to-parts warehouses are not strictly outdated and can still be the right choice today. Warehouses for spare parts in the aircraft industry are a good example. To reduce expensive ground time of maintained aircraft, they mainly have to provide many large and individual spares under great time pressure. These product characteristics impede both the batching approaches of the 2nd generation (i.e., large and urgent products tend to require individual tours anyway) and the robotized handling of the 3rd generation (i.e., large and damageable SKUs still defy automation).

3.2. 2nd generation

System setup: Among the most wide-spread extensions of the basic 1st generation setup of picker-to-parts warehouses are certainly *batching* and *zoning*. The former allows to unify multiple customer orders to a larger pick list, which is then jointly processed in an extended picker tour. In this way, the pick density per tour is improved and unproductive travel is reduced. Under a zoning policy, the warehouse is partitioned into distinct zones each operated by a dedicated picker workforce. This reduces the area to be traversed by each picker, eases the pickers to familiarize with their purview, and allows to extend the potential storage area from which demanded SKUs can be obtained. Zoning can be applied in a *parallel* or *progressive* manner (see also De Koster et al., 2007). Parallel zoning promises a speedup of order completion, because it allows multiple zones to start order picking on the same subset of orders in parallel, each in their separate zone. Progressive zoning, instead, sends a dedicated order bin subsequently from zone to zone (typically via a conveyor), where each zone adds the ordered products in its area. Hence, this system is also denoted as pick-and-pass (De Koster et al., 2007) or sequential zoning (Van der Gaast et al., 2019). The subsequent zone visits of progressive zoning prolong order completion. On the positive side, however, orders are readily sorted into dedicated bins and can be prepared for shipping without further sortation effort.

Batching and parallel zoning, instead, require an extra consolidation stage, where batched orders are separated and/or partial orders arriving from multiple zones are unified. A fully-automated crossbelt sorter, which is one of the order consolidation tools frequently applied in real-world warehouses, is depicted in Fig. 2. This extra consolidation stage, typically coming along with significant investment and operational costs, is the main disadvantage of batching and parallel zoning (see also Boysen et al., 2019a).

Another extension that has been innovated by large e-commerce warehouses is *scattered storage*. Alternative denominations are mixed-shelves, dispersed, or even explosive storage. While the general possibility to store a SKU on a handful of different locations has been considered in warehousing research for quite a while (e.g. Daniels et al., 1998), scattered storage runs this general idea to the extremes. Unit loads of homogeneous products are broken up and individual pieces are stored all over the different zones and shelves of a warehouse. This leads to the widely mixed shelves depicted in Fig. 3(middle). Especially in e-commerce, average order sizes are small and assortments are huge (Boysen et al., 2019b). Hence, it is hardly predictable what SKUs end up jointly on the same pick lists. Whatever it is that is ordered jointly, however, under the scattered storage policy there is a good chance that somewhere in the warehouse these products are stored closely together. The price for the resulting reduction of unproductive picker travel is a more involved replenishment process. Instead of just placing a complete unit load into storage (e.g., with a single forklift move), replenishment workers (often called *stowers*) must traverse the warehouse in order to place individual pieces into open shelf spaces (see also Weidinger & Boysen, 2018).

Note that all three adaptations (i.e., batching, zoning, and scattered storage) are merely organizational extensions of the 1st generation setup, so that this generation distinction is rather a structural choice but not based on technological progress.

Layout design: In addition to the decisions already related to 1st generation of picker-to-parts warehouses, the main additional layout design choices for 2nd generation systems concern (a) the number and placement of zones, if the zoning policy is applied, and (b) the choice and setup of a suitable sortation device.

(a) Number and placement of zones: Naturally, a main influencing factor on the layout design decisions is whether parallel or progressive zoning is applied. More zones increase the division of labor under parallel zoning and thus speed up the retrieval of the demanded products. On the negative side, however, more zones also tend to increase the number of bins, in which products of a specific order are contained. If only one of these bins gets delayed (e.g., by a conveyor failure), then all other orders that have already arrived at the consolidation stage (and block capacity there) get delayed. In progressive zoning, more zones reduce the operating area of each zone's workforce, but also increase the average number of zones visited by each order. This delays order completion, because especially when entering new zones significant waiting times might occur. Naturally, these trade-offs are not only impacted by the number of zones but also their placement on the shop floor and the storage assignment of products to zones. To account for this interdependence, Gray et al. (1992) present a hierarchical decision framework that involves all these decisions, Petersen (2002) focuses on the placement and sizing of zones in an extensive simulation study, and Van der Gaast et al. (2019) formulate an extensive queuing network model that involves all relevant decisions.

Instead for applying fixed zones, however, it is also possible to dynamically vary the zone sizes according to the current workload. An appealingly simple idea to achieve this without a continuous replanning is the *Bucket Brigade* approach introduced by Bartholdi and Eisenstein (1996). To organize the division of labor among multiple pickers who accompany order bins moved on a manual roller conveyor along a straight pick face, the basic protocol of this approach only consists of two rules. Forward part: Move onward with the current order bin and pick all demanded products, until you meet your successor. Then switch to the backward part: Walk back until you meet your successor and take over his or her order bin. Continue with the forward part. Bartholdi and Eisenstein (1996) show that the zone sizes operated by each picker under this protocol are self-balancing, so that they adapt automatically to workload changes. In Bartholdi et al. (2001), a real-world implementation in a distribution center is reported, and an in-depth survey on the manifold research on Bucket Brigades is provided by Bratcu and Dolgui (2005).

(b) Choice and setup of a sortation device: If the application of batching and/or parallel zoning demands an additional consolidation stage, then further layout decisions are related to the sortation device. In a first stage, it must be decided whether an automated or a manual solution is better suited to consolidate the incoming products according to customer orders. Next to the crossbelt sorter depicted in Fig. 2 there are other automated solutions, such as bomb-bay, sliding-shoe, or tilt-tray sorters (see Boysen et al., 2019a). They only vary in the technical device that redirects products from the main conveyor into the packing lanes where orders are collected. Since sortation is only required for multi-piece orders that still largely defy fully-automated packing processes, the packing lanes are typically manned with human workers putting the orders into their packaging. A slightly different automated sortation device is a pocket or bag sorter, where pieces are transported in small bags hanging from the rails of an overhead conveyor (see Boysen et al., 2023a). A manual sortation solution, instead, is based on a so-called put wall (see also Boysen et al., 2019c). On one side of the wall, which is merely a person-high rack with many reach-through shelves where orders are collected, there are the *putters*. They place incoming products into shelves. Guided by a light signal that identifies completed orders, the other side of the wall is manned by the packing workforce. The research question under which circumstances either automated or manual order consolidation shows to be advantageous is, for instance, treated by Petersen (2000) and Russell and Meller (2003). A further important layout decision related to automated sortation devices concerns the right number of packing lanes, which is investigated by Bozer and Sharp (1985).

Storage assignment: If zoning is applied, then the basic storage assignment policies of the 1st generation picker-to-parts warehouses can still be applied within each zone. Beyond that, however, it is also necessary to partition the total assortment of SKUs among zones. This decision task faces the additional requirement that the total workload should be equally distributed among zones (Brynzér & Johansson, 1995). Unbalanced solutions produce bottlenecks, increase waiting times within the other zones, and lead to unfair divisions of labor among the picking workforce. Simple planning approaches to avoid this for parallel zoning are, for instance, presented by Yuan et al. (2018). Note however, that "balance" obtained based on deterministic assumptions may not perform as expected in a stochastic setting in progressive zoning, where both the work content of orders per zone varies and workers may work at different speeds. This explains why in bucket brigade picking the largest throughput can be obtained by sequencing pickers from slow to fast (Bartholdi et al., 2001). When speeds are variable, the throughput-maximizing worker sequence depends on both the mean and standard deviations of the worker speeds (Bukchin et al., 2018).

In a scattered storage warehouse, a storage assignment solution decides on the assignment of individual products to open shelf spaces in the racks. In many real-world warehouses, this assignment is not optimized but left to the individual choice of the *stowers* (Boysen et al., 2019b). Analogously to their colleagues from the picker workforce, stowers receive bins filled with products at an I/O point of the central conveyor system, load these bins on manual picking carts, and start their travel through the warehouse. They are advised to stow products according to their individual choice whenever they pass by open shelf space. Analogously to closest-open-location storage (see Section 3.1), this simple protocol supports closely packed shelves near the I/O points. On the negative side, however, it is barely possible for stowers to consider how many pieces of the same SKU are already stored in the vicinity of a currently open shelf space. Hence, clusters with many pieces of the same SKU in a confined area of the warehouse can arise, which violates the basic principle of scattered storage. To avoid this, Weidinger and Boysen (2018) introduce an optimization approach for storage assignment that maximizes the "scatteredness" of products. Specifically, they minimize the weighted sum of maximum distances toward the closest piece of each SKU seen from the I/O points. Similar approaches with alternative measures for the dispersion of products are, for instance provided by Zhang et al. (2020). Further research on the best "measure of scatteredness" and the gains compared to the unoptimized status quo is, however, required.

Order processing: The batch formation problem, i.e., the decision which picking orders should be jointly processed per picking tour, is another classic from warehousing research. The survey papers of Henn et al. (2012), Cergibozan and Tasan (2019), as well as Pinto and Nagano (2022) document the manifold research in this area in a comprehensive manner. Existing research mainly focuses on the basic trade-off between efficiency and urgency. Efficiency suggests to unify especially those orders whose picking positions are in neighboring areas, so that the total travel effort of pickers is reduced. The flexibility to identify suitable orders to realize this aim is, however, reduced if the urgency of orders reduces the freedom to postpone their processing.

In the most basic version, this trade-off can be resolved by assuming that the most urgent orders have already been selected on a previous decision stage. Hence, only the minimization of the total travel effort for the given order set remains. The recent paper of Aerts et al. (2021) shows that the remaining problem is equivalent to the clustered

vehicle routing problem. Strong NP-hardness of the problem, even in the parallel-aisle structure of warehouses is proven by Gademann and Van de Velde (2005). Alternatively, orders can also explicitly be assigned due dates, derived from the delivery dates promised to customers, in order to minimize some due date-oriented objective (e.g., see Scholz et al., 2017). Next to these two research streams on static and deterministic batching, there is also research on stochastic dynamic problems. Driven by increasing time pressure, e.g., in dark stores where within-the-hour deliveries are promised (see Section 1), it may be necessary to quickly decide whether a new order that has just popped up should be integrated into an ongoing tour. A good example for this stream of research is provided by Bukchin et al. (2012), who model the decision whether a picker should depart with the current batch onto the next tour or wait for further urgent orders as a Markov decision process. Finally, another stream of batching-related research extends static and dynamic problems by further peculiarities of practical relevance. Examples are congestion avoidance among multiple pickers operating their batches in parallel (Hong et al., 2012), the consideration of different processing speeds of the picking workforce (Matusiak et al., 2017), and the integration of returned products into the batching process (Wruck et al., 2013).

Order processing in a scattered storage warehouse extends the classical picker routing problem elaborated for 1st generation picker-to-parts warehouses in multiple ways. Foremost, since each product defined on a pick list is typically available at alternative shelf positions, the selection from which shelf position each product is picked becomes part of the problem. Weidinger (2018) shows that this additional decision task renders the classical picker routing problem strongly NP-hard even in a parallel-aisle warehouse. Further peculiarities are often caused by multiple I/O points to the conveyor systems, where the bins carried on the picking cart can be handed over in a dynamic manner. The latter allows that not all bins carried on a cart are handed over at the same I/O point but at different ones whenever a subset of bins is completed. Furthermore, pickers can park clumsy carts in the warehouse in order to obtain a handful of products without obstruction on *cartless subtours*. Optimization approaches for these extended picker routing problems suitable for real-world scattered storage warehouses have been provided by Weidinger et al. (2019) as well as Schiffer et al. (2022).

Another frequently cited number in warehousing research is the 80% of all warehouses of Western Europe that still follow the picker-to-parts setup (Napolitano, 2012). Given the clear trend toward automation, warehousing research should certainly reevaluate whether this figure is still up to date. Even if it has declined, however, warehouses with picker-to-parts systems of the 1st and 2nd generation still constitute the vast majority. Opening up these existing warehouses to automation without having to completely remodel (or even rebuild) them according to the parts-to-picker paradigm leads us to the currently evolving 3rd generation picker-to-parts warehouses of the following section.

3.3. 3rd generation

System setup: The main functional steps in a picker-to-parts process are the travel through the warehouse and the product retrieval from the shelves. Modern 3rd generation picker-to-parts warehouses either automate both of these steps or just the former with the help of autonomous robots. Autonomous mobile picking robots, such as the Toru robot of Magazino depicted in Fig. 2, move autonomously through the warehouse. Furthermore, they are equipped with an automated gripping device and a storage bay to directly retrieve products from the shelves and to intermediately hold picked products, respectively. This allows them to completely substitute human pickers.

Despite the significant progress in automation in the recent years, however, especially gripping (e.g., with a vacuum gripper) is still a huge challenge in robotics (e.g., see Correll et al., 2016). Hence, the Toru robot, for instance, is yet restricted to rectangular packagings (e.g., shoeboxes) and does not meet the performance of human pickers. Another 3rd generation solution, thus, only consigns the transportation task to autonomous mobile (transport) robots (AMRs, see Fragapane et al., 2021), whereas human pickers are still in charge of the product retrieval. Humans pick requested products from the shelves and put them into order bins, which are transported by AMRs. In this way, humans and robots cooperate during the picking process. Once a pick list is completed, the respective AMR autonomously delivers all picked products back to the depot. The picker, instead, stays in the storage area and meets with the next AMR to proceed with the next pick list. In this way, human pickers are relieved from the (unproductive) travel back and forth the depot, which promises a more efficient picking process. Note that there are prototypes (e.g., the fetch and freight concept of Fetch robotics) that propagate the separation of product retrieval and transport even if both tasks are executed by two different types of mobile robots.

The impact of (a) fully autonomous picking robots and (b) AMR-assisted picking on the picker-to-parts process and its decision problems is elaborated in the following.

Layout design and storage assignment: The main decision tasks regarding the layout and the storage assignment remain the same as elaborated for the 1st and 2nd picker-to-parts generation: We still must decide on the placement of the racks on the shop floor and the storage positions of products in shelves. Whether (and if how) these well-established decision tasks are impacted by the application of (a) fully autonomous picking robots and (b) AMR-assisted picking is still unclear. Regarding (b), Boysen et al. (2019b), for instance, conjecture that fast-moving products should rather be stored in the middle of the warehouse, because this is the main activity area of human pickers assisted by AMR. Löffler et al. (2023), instead, show that having the fast movers closer to the depot improves AMR-assisted picking performance. A final resolution of this contradiction and exploring all other layout- and storage assignment-related decisions still demands a lot of future research.

Order processing: If (a) fully autonomous picking robots substitute human pickers, the picker-to-parts process itself remains mostly unaffected. One important difference, however, is implied by the storage bays into which picking robots load their picked products. If a storage bay is full, then the robot autonomously returns to the depot, where order racks are positioned into which robots automatically hand over the demanded products from their storage bays. There is, however, no technical requirement that all products of an order must be delivered by the same robot into the respective order rack. Picking of an order can, thus, be split among multiple robots, which increases flexibility and promises a more efficient picking process. There is yet not much research on this issue (among the very few exceptions is Jiang et al., 2021).

Order processing in (b) an AMR-assisted order picking environment, is strongly influenced by the policy that regulates the cooperation among humans and robots. The most restrictive policy is what Löffler et al. (2023) call the *fixed-assignment* policy. Under this regime, a picker and an AMR build a fixed pair while processing a specific pick list. Once a pick list is completed, the AMR returns to the depot and the picker pairs with another AMR for the next pick list. Löffler et al. (2022) show that under this policy (and further assumptions, e.g., pickers need not wait for AMRs when pairing) the picker routing problem becomes equivalent to the clustered TSP. They further show that the clustered TSP is solvable in polynomial time in the parallel-aisle structure of warehouses, if the cluster sequence is given. A more flexible cooperation is enabled by the *free-floating* policy (Löffler et al., 2023). Here, any picker can satisfy any AMR demand, defined by the pick list assigned to the respective AMR. The increase of flexibility, however, comes at the price of a very challenging holistic scheduling problem where all pickers and AMRs need to be synchronized. Löffler et al. (2023) show that such a system is prone to ripple effects where delays of one pick may propagate through the warehouse and influence all other pickers and AMRs. More research on this effect and how to overcome it is required.

3.4. Future research needs

Despite the long tradition of picker-to-parts systems and their evolution through three generations, we still see more research work to be done. Some of these future research needs have already been elaborated in the prior sections. Beyond the various issues listed there, we see two decision tasks in particular that demand special attention:

(a) Batching: Our elaboration in Section 3.2 shows that batching (i.e., the unification of multiple customer orders to a larger pick list and their processing on a joint picker tour) attracted a lot of research. The vast majority of previous research, however, (in one way or another) focuses on the basic efficiency-urgency trade-off. Recall that the flexibility to batch suitable orders with close-by storage positions increases, if a less strict time regime allows a more generous order postponement and thus widens the set of orders to choose from (and vice versa). This is without doubt an elementary trade-off related to the batching decision; but in many warehouses it is not the only aspect to be considered. Especially in large e-commerce warehouses, batching is often combined with parallel zoning (see Section 3.2). In such a setting, it may occur that bins containing partial orders of the same batch must wait in intermediate storage for a delayed bin from one zone. Situations like this become more likely, the more bins and zones are involved in processing a batch of orders. If the complete intermediate storage capacity is taken, a so-called gridlock (Gallien & Weber, 2010) threatens: All batches whose bins wait in intermediate storage miss some products and cannot be further processed, but there is no space in intermediate storage to add additional bins that deliver the missing products. Resolving such a deadlock requires excessive manual work and must, thus, be avoided. Hence, how to distribute batch processing among multiple zones is an important additional decision task that has widely been neglected. Especially in scattered storage warehouses, where products may be available in multiple zones, this decision is crucial. New batching models are thus required that (i) reduce the gridlock risk, (ii) aim at a fair division of labor among the different zones, (iii) include the additional flexibility of scattered storage, and (iv) consider the impact on the subsequent consolidation stage. The latter means that batches, completed in the picking stage, should continuously arrive at the consolidation area, so that they neither starve for work nor become overloaded.

(b) Furthermore, routing procedures that optimize the *replenishment process for scattered storage* warehouses constitute another challenging research task that has yet been neglected. Most scattered storage warehouses (we have visited) leave the storage assignment decisions to the stower workforce. They must decide without decision support where to place the products from their carts into open shelf space. The storage assignment decision must trade off the scatteredness of products (i.e, pieces of the same SKU should be equally spread among shelves) and the stowing effort (i.e., the picking tour along the selected stowing positions should be short). Optimized routing procedures can help streamlining this process, especially if during peak times open shelf space is scarce. Moreover, it should be considered that the stower workforce competes for open shelf space, so that the open storage positions must be coordinated among the stower team. This constitutes new challenging picker routing problems that have not been properly addressed by warehousing research yet.

4. Parts-to-picker systems

Parts-to-picker systems reverse the conventional picker-to-parts approach, with the aim of reducing unproductive travel for pickers. In these systems, SKUs are delivered to stationary pickers, who may be humans or automated picking devices, operating at dedicated picking workstations. Although this eliminates the need for pickers to traverse the storage area, it requires a larger upfront investment in hardware. Parts-to-picker systems typically comprise three essential components. Firstly, there is a storage system equipped with a storage and retrieval machine. These systems commonly consist of racks designed to hold either bins, totes, or pallets. These loads can be homogeneous unit loads or heterogeneous containers filled with various SKUs. Secondly, one or more workstations are set up where the requested products are retrieved. Lastly, a conveyor system connects the aforementioned elements.

The initial storage and retrieval machines were primarily forklifts operated by humans. However, over time, they have been increasingly replaced by automated cranes and other automated storage and retrieval machines. Forklifts have a rich history spanning approximately a century and have played a pivotal role in transforming material handling and warehousing practices. They have enabled the efficient utilization of vertical space by storing unit loads in racks and facilitated the transportation of larger loads over greater distances. Automated cranes represent a significant advancement in this field. They are capable of automatically storing and retrieving loads within high-bay racks up to a height of 40 m. The origins of automated cranes in high-bay warehouses can be traced back to the 1960s, with Demag being credited as one of the pioneers.

In the 1st generation of parts-to-picker systems, the workstations are supplied with SKUs by rack-based storage systems operated by an automated crane (or forklift). In the 2nd generation novel storage systems, such as deep-lane and shuttle-based systems, are applied, whereas the 3rd generation is based on robotized retrieval systems. Typical representatives of these three generations are depicted in Fig. 2. We discuss these three generations of parts-to-picker systems in the following sections.

4.1. 1st generation

System setup: Automated parts-to-picker systems of the 1st generation, based on crane-operated storage systems, are commonly referred to as *AS/R* (Automated Storage and Retrieval) systems. They typically consist of aisle-captive cranes that can pick-up a load from an I/O point, usually a buffer position on a conveyor, and store it in shelves of racks positioned to the left and right of the crane aisle. The cranes are connected to two rails: one on the floor and another one mounted to the top of the racks. This gives the cranes stability and allows them to drive and lift simultaneously, at relatively high speed. The system typically serves as a separate silo building, as the racks are cladded with side panels and the roof is connected to the racks. Next to retrieval or storage in single-command cycles (i.e., processing a load by moving it from the I/O point to a storage location and returning empty), it is also possible to carry out so-called dual command cycles, wherein a storage request is combined with a retrieval request, which saves one crane movement per dual command cycle. Systems that store bins or totes are also called a *miniload*. After handing over a load at the I/O point, it is directed to a workstation via a conveyor system, where the required quantity is picked for a customer order. Finally, the load returns to storage following the reverse process.

Alternatives to this most common system setup are detailed in the in-depth survey papers on AS/R systems provided by Roodbergen and Vis (2009), Gagliardi et al. (2012), and Boysen and Stephan (2016). For instance, there are AS/R systems with aisle-changing cranes (or forklifts), multiple interfering cranes per aisle, and multi-shuttle cranes that allow the simultaneous transport of multiple loads. In this paper, we focus on the most basic 1st generation setup without these potential extensions.

Layout design: The main decisions to be taken during the layout design phase relate to the number of workstations as well as the number and size of storage aisles to supply them. Determining the number of workstations is comparatively simple; in the most basic approach only the desired total order throughput is divided by the average order throughput per station. However, deriving the right number and size of storage aisles to supply the resulting workstations without producing a bottleneck is more complicated. Unfortunately, the throughput of a

specific layout depends not exclusively on the layout choices but also on the subsequent decision areas related to storage assignment and order processing.

To avoid the excessive effort for deriving simulation models for each combination of layout alternative, storage assignment, and operational policy, TT models (see Section 2) are thus a suitable tool to estimate the performance of different layout designs in a straightforward manner via closed-form solutions. Bozer and White (1984) were among the first to study TT models for 1st generation parts-to-picker systems. They calculate the single and dual command cycle time of a crane, for racks with random storage (any destination location within the rack is equally likely to be selected) and the I/O point at a bottom rack corner. These formulas can be used to calculate the optimal length-to-height ratio of an aisle. This proves to be *square in time*, meaning the times to reach the farthest and the highest location are identical. Since a crane travels about four times faster than it lifts (Boysen & Stephan, 2016), the aisle's physical length to height ratio should be about four to one to minimize the cycle travel time. Given the demanded storage space and the throughput per hour, these formulas can finally derive the demanded number of aisles (Bozer & White, 1984).

Storage assignment: When deciding on the storage positions of the loads within the storage system there are basically the same policies as already discussed for picker-to-parts systems: (a) random storage, (b) closest-open-location storage, (c) dedicated storage, (d) full-turnover storage, and (e) class based storage. Note that a family grouping is less relevant, especially for single-shuttle cranes restricted to one load at a time. The above TT formulas have been extended to handle these alternative storage policies, but also further aspects such as crane acceleration and deceleration, different locations of the pick-up and delivery I/O point, discrete racks, and multi-shuttle cranes.

An important impetus for cycle time calculations with class-based storage was given by the seminal work of Hausman et al. (1976). They calculate the single-command cycle times and optimal class boundaries for given ABC demand curves. Their calculation accounts for product replenishment according to a continuous review (r, Q) reordering policy. Their results are extended for any number of storage classes by Rosenblatt and Eynan (1989). However, these research works do not consider that with fewer storage classes more SKUs are stored in each class, which requires less space. The latter is true because loads can share the space within their class, which reduces if stocks of more SKUs are replenished at different points in time. This means that an optimum number of storage classes exists, which turns out to be small in most real-world warehouses (about 3 to 5) (Yu et al., 2015).

Order processing: Once the layout and storage assignment decisions are taken and a given set of customer orders must be processed, there are two different types of operational decision tasks to be solved: (a) crane scheduling in the storage system and (b) the synchronization of order processing and load delivery at workstations. Instead of giving either the storage system or the workstations priority (and fixing the other task according to the planning result for the prioritized element) both decision task can (c) be planned in an interdependent manner.

(a) Crane scheduling decides on the sequencing of storage and retrieval requests for a given set of loads and their unification to command cycles, such that the total travel time of the crane is minimized. This (or a related crane-oriented) objective is intended to relieve a bottleneck storage system. Lee and Schaefer (1997) were the first to show that pairing storage and retrieval requests to dual commands for loads with dedicated storage positions and a single I/O point can be solved in polynomial time via the linear assignment problem. Further crane scheduling problems solvable in polynomial time have, for instance, been identified by Van Den Berg and Gademann (1999) (separate I- and O-points at both ends of the rack and a given sequence of storage requests) and Gharehgozli et al. (2017) (two I/O points at both ends). For an in-depth survey reviewing these results and providing a classification scheme for the crane scheduling problems of AS/R systems refer to Boysen and Stephan (2016).

(b) Workstation synchronization can improve the order throughput if customer orders are processed simultaneously in a workstation. In this case, orders, processed at the same time and demanding the same SKU, can be serviced by the same load. This saves storage and retrieval requests, so that a bottleneck AS/R system can be relieved and the order throughput is improved (Füßler & Boysen, 2019). Boysen et al. (2023b) provide an in-depth review of the family of slightly varying workstation synchronization problems. They show that this decision task can provide a more significant relieve of a bottleneck AS/R system than crane scheduling.

(c) Naturally, the storage system and the workstations are not independent. Therefore, it is advisable to address both of the preceding operational decision tasks (a) and (b) in an *interdependent manner*. Research in this area, however, is rare. Among the very few exceptions is the paper of Baardman et al. (2021). They consider an interdependent crane scheduling problem where loads, retrieved to visit a workstation, become storage requests after progressing through the workstation's load buffer. Thus, dual commands cannot be determined independently. Boysen et al. (2022b) build up on this research and show that the resulting optimization problem is strongly NP-hard but is conveniently solvable via a transformation to the standard vehicle routing problem.

It is our impression that the majority of existing parts-to-picker systems still follow the 1st generation parts-to-picker setup. Other than picker-to-parts systems, however, where 1st generation systems can still be the right choice today (see Section 3.1), most newly erected parts-to-picker system no longer apply a crane-operated AS/R and rather use 2nd or 3rd generation systems.

4.2. 2nd generation

System setup: To improve the 1st generation process, the 2nd generation of parts-to-picker systems uses alternative storage systems. We concentrate our survey on the most important ones. For a discussion of less frequent 2nd generation systems, such as (horizontal and vertical) carousels, vertical lift modules, and puzzle-based storage systems, refer to the literature instead (e.g., Azadeh et al., 2019a; Litvak & Vlasiou, 2010; Meller & Klote, 2004).

To improve the storage density of 1st generation systems, crane-based AS/R systems can also store multiple loads directly behind each other in the same lane. They are then called *(a) deep-lane AS/R systems*. Many of these systems store loads either double deep or 1.5 loads deep. In the latter case, three loads are stored in two adjacent racks and the middle load can be accessed from two neighboring aisles. The cranes are then equipped with telescopic forks, and the foremost storage positions must be empty to access a load of a deeper position. If even deeper storage is required, cranes can be equipped with a fixedly connected shuttle platform (called satellite) that can be released into a lane while the crane waits. Multi-deep systems are widely used in refrigerated and frozen storage, where space savings immediately translate in reduced cooling cost. To avoid blockings (i.e., loads with later retrieval time block demanded loads stored behind them in the same lane), it is aspired to only store "identical" loads, e.g., pallets of the same SKU or goods for the same recipient, in the same lane.

(b) Shuttle-based storage (SBS) systems: Other than cranes, which allow for a simultaneous vertical and horizontal movement, SBS systems separate these movements and apply independent lifts and shuttles instead. Shuttles are restricted to the movements back and forth on the same rack level. Typically, each rack level in an aisle has its own shuttle (tier-captive systems). But there are also tier-roaming systems, in which the shuttles can be relocated between tiers and aisles. Vertical lifts exchange loads with the shuttles and move loads between a rack level and the I/O point. In the case of roaming shuttles, lifts also move shuttles between the storage levels. The loads to be retrieved are mostly bins and totes, stored in single deep racks. Shuttle-based systems have gained popularity, particularly in e-commerce retail. Compared

to crane-based AS/R systems, they offer a much higher throughput capacity (due to the division of labor) at lower cost (due to the use of only standardized low-cost components).

Layout design: (a) The outstanding layout decision added by deep-lane AS/R systems (compared to their single deep ancestors) relates to the question how deep the storage lanes should be. Too deep racks may lead to *honeycombing* (Tompkins et al., 2010): if there are not enough identical loads to completely fill a lane, parts of them must remain empty to rule out blockings. This leads to low average lane fill rates. Undeep racks require multiple lanes per identical load, which demands more lanes and thus longer aisles as well as increased crane travel times. To resolve this trade-off, Goetschalckx and Ratliff (1991) suggest a solution approach based on an adapted TT model and dynamic programming.

(b) The main layout decisions related to SBS systems concern suitable dimensions, determined by the number and length of lanes as well as the number of tiers. To evaluate a specific layout where these decisions are taken, the separation of horizontal and vertical movement alters the TT models to estimate the throughput. Malmborg (2002) was among the first to derive TT models for single and dual commands of SBS systems. Extensions including acceleration, deceleration, and the maximum shuttle velocity are provided by Lerher et al. (2015), whereas Roy et al. (2012) base their layout evaluation on semi-open queuing networks. The latter have also been applied to evaluate further layout choices, such as the positioning of the lift (Roy et al., 2015).

Storage assignment: (a) The storage assignment decision of deep-lane AS/R systems has to handle the additional complexity of potential blockings among different loads sharing the same lane. For a long-term assignment of SKUs based on dedicated storage, eliminating blockings is straightforward. Therefore, existing research mainly focuses on short-term storage of loads prepared for the pickup of delivery vehicles. To account for uncertain arrival times of the delivery vehicles, Zaerpour et al. (2015) assign prospective retrieval intervals to each load, so that only loads with non-overlapping intervals are allowed sharing the same lane. Their research is extended by Boysen et al. (2018) for two-sided lanes, where it is enough that one side of the lanes is not blocked.

(b) Analogously to crane-based AS/R systems, SBS systems aim to store frequently demanded SKUs closer to the lift in order to reduces handling times. Adaptions of TT models, evaluating class-based storage for single and dual commands, are provided by Dong and Jin (2021). An evaluation of class-based storage based on queuing networks is provided by Roy et al. (2012).

Order processing: (a) During the retrieval process from deep-lane AS/R systems, the resolution of blockings constitutes the outstanding challenge. If demanded loads are not directly accessible, the blocking loads must be reshuffled into other lanes, where they could produce further blockings. Minimizing the number of reshuffles in these cases is therefore treated by Dong et al. (2021) and Yang et al. (2023).

(b) The operational scheduling of storage and retrieval requests in SBS systems requires a complex synchronization between lifts and shuttles. Fast heuristic optimization procedures for this task are, for instance, provided by Carlo and Vis (2012) (priority rule with look ahead), Wang et al. (2015) (genetic algorithm), and Chen et al. (2023) (adaptive large neighborhood search).

4.3. 3rd generation

System setup: The 3rd generation aims to remove a major disadvantage of parts-to-picker systems: the inflexible structure once a system is erected. The lack of flexibility is caused by the abundance of fixedly installed hardware. In 3rd generation systems, at least parts of this hardware are substituted by AMRs, which can be added to and removed from a system in a more flexible manner. Specifically, we discuss robotized parts-to-picker systems based on (a) shelf-lifting, (b) grid-supported, and (c) rack-climbing robots.

(a) Shelf-lifting robots: In robotic mobile fulfillment systems, the shelves (also denoted as inventory pods) are not fixedly installed but can be lifted and delivered to workstations by shelf-lifting mobile robots (see Fig. 1(d)). Without a load, these AMRs can travel underneath the pods; but to transport a load, travel aisles are needed. At the workstations, human pickers retrieve the demanded products from the pods, which typically follow a scattered storage policy and contain a mix of different products. After picking, an AMR returns a pod into the storage area, where it waits for further pickup to one of the workstations. The basic concept of shelf-lifting AMRs dates back to Jünemann (1989). It was first introduced on the market by KIVA Systems, which became Amazon Robotics after its acquisition. Nowadays, Amazon is using this concept in all its new warehouses (Allgor et al., 2023), and also other e-commerce retailers have followed (Qin et al., 2022). Many other system suppliers are now offering shelf-lifting robots in combination with mobile racks.

(b) Grid-supported robots: Parts-to-picker systems based on grid-supported robots store products in bins stockpiled on top of each other (see Fig. 1(b)). The resulting stacks of bins are supported by a grid-shaped frame on which AMRs, each equipped with a lifting mechanism, travel autonomously. If a demanded bin is blocked by other bins, they must be relocated to other stacks. A retrieved bin is transported by AMR toward a lift that connects the workstations at the periphery of the system. Autostore (www.autostoresystem.com) is the best-known supplier of these systems, but there are competitors (e.g., Jungheinrich Powercube, Intellistore, and Ocado) with similar offers.

(c) Rack-climbing robots: Only recently, rack-climbing robots like those offered by Opex, Exotec (see Fig. 2), and Attabotics have emerged. These systems do not require lifts or cranes for vertical transport in the storage system; the AMRs can climb the racks autonomously. Hence, they can access all storage positions without much interference. Exotec robots can not only climb the racks but also travel on the floor and bring a bin, retrieved from the storage system, directly to a workstation.

Layout design: The main decision task added by all three AMR-based systems, on top of the typical layout decisions concerning the size of the storage system and the number of workstations, is the sizing of the robot fleet. More AMR promise an improved throughput but also lead to more traffic (and thus congestion). The latter is also influenced by the layout of the storage system and the resulting interference of AMRs on the driving lanes, so that a concerted planning of all layout issues seems advisable. Queuing networks for this task are introduced by Lamballais et al. (2017), Zou et al. (2018), and Azadeh et al. (2019b) for systems based on (a) shelf-lifting, (b) grid-supported, and (c) rack-climbing robots, respectively.

Storage assignment: All three robot-based systems apply completely different types of storage devices, whose impact on the storage assignment decision is briefly elaborated in the following. (a) Shelf-lifting mobile robots induce two interdependent storage assignment decisions. Firstly, products must be stowed into inventory pods, and, secondly, pods must be parked somewhere in the storage area. Decision support for the former decision that aims at an increase of the mix of SKUs stored in scattered storage pods is provided by Kim et al. (2020). The rather short-term decision for the right parking spots must position frequently demanded pods closer to suitable workstations, which is, for instance, investigated by Weidinger et al. (2018). (b) Since bins are frequently relocated by grid-supported robots, storage assignment is no longer a long-term decision task in these systems. Instead, the storage assignment is the dynamic result of multiple operational decisions on the stacks blocking bins are relocated to. Furthermore, off-peak times can be utilized to reshuffle the stacks. Decision support for these operational storage assignment decisions is yet missing. (c) Systems based on rack-climbing robots apply a traditional rack-based storage system, where the typical storage assignment policies elaborated in Sections 3.1 and 4.1 can be applied. To adapt the evaluation of class-based and random storage for these systems, however, Chen et al.

(2022) include the AMRs' bin delivery toward the workstations into their analysis, based on semi-open queuing networks.

Order processing: (a) On an operational level, orders processed with the help of shelf-lifting robots must be assigned to workstations, and, to satisfy the resulting demands of a workstation, pods must be selected and sequenced. Furthermore, the transport of pods toward workstations must be assigned to AMRs, whose path finding on the shop floor must be coordinated in real-time. Boysen et al. (2017) were the first to schedule the order processing by an interdependent sequencing of orders and pods at a single workstation. Numerous extensions, e.g., integrating multiple workstations (Zhuang et al., 2022), split orders (Xie et al., 2021), and a dynamic adaption to varying picking performance (Sheu & Choi, 2023), have been published since then. (b) In systems based on grid-supported robots, orders must also be assigned to workstations. Furthermore, a bin containing a demanded product must be selected, all selected bins must be sequenced, transport jobs must be assigned to robots, and the stacking decisions for reshuffled and returning bins must be made. All these highly interdependent operational decisions leave plenty of room for future research. Existing research mainly provides simulation studies (e.g. Beckschäfer et al., 2017), whereas research applying more sophisticated methods is yet missing. (c) Also for systems based on rack-climbing robots, the operational decisions (i.e., the assignment of orders to workstations, bin selection and sequencing, as well as the assignment of transport jobs to AMRs) have not been considered yet and provide a fruitful field for future research.

4.4. Future research needs

The three generations of parts-to-picker systems have attracted a massive body of scientific research. Nevertheless, we still see many open research challenges. Some of them have been documented above when discussing specific parts-to-picker systems, but we also see the following general trends related to parts-to-picker systems.

(a) Right storage system: Our above elaborations document a huge difference in the storage systems and their specific storage and retrieval mechanisms. A lot of scientific effort has been invested to explore the capability of each individual system, but a systematic cross comparison is yet missing. A structured benchmark test could deliver valuable decision support for practitioners regarding the right storage system choice depending on factors such as order structure, assortment, and lead time.

(b) Setup of workstations: The workstations have evolved significantly over the centuries. Modern stations can be adapted ergonomically to individual picker needs and contain an automated weighing mechanism of bins to warn against picking errors. Latest workstations are even fully automated, e.g., by an industrial robot equipped with a vacuum gripper. The performance impact of these workstation extensions has not been systematically benchmarked in a scientific manner. Also the question on the right number of customer orders, processed in parallel on the workbench, deserves attention. More bins increase the synchronization effect, so that multiple orders demanding the same SKU can be serviced jointly by the same loads, but they also increase the picker's unproductive movement along an extended workstation.

(c) Holistic research including storage system and workstations: The vast majority of parts-to-picker research exclusively focuses on a specific storage system and neglects the impact on the workstations. Maximizing the throughput of a storage system or minimizing the makespan for a given set of storage and retrieval requests, however, neglects the picking times in the workstations. Processing each single bin in a workstation takes a specific amount of picking time, which leaves the storage system time to deliver the next bin. An even faster bin delivery before the picker has finished the previous bin, however, does not speed up the complete process. Only if bin delivery exceeds the picking time, a workstation starves for work and the order throughput deteriorates. Thus, the interdependence between storage system and workstations, including the buffer provided by the interconnecting conveyor system, certainly deserves more scientific attention to derive more realistic warehousing models.

5. On the future of warehousing research

This section explores future research opportunities that extend beyond the scope of picker-to-parts or parts-to-picker systems, aligning with broader warehousing trends. We begin by addressing three topics that have previously garnered attention as noteworthy areas for future research, which have since attracted substantial research efforts. While acknowledging the significance of research in these domains, it is essential to recognize their limitations. Consequently, we caution against an excessive focus on these subjects, advocating for a balanced approach. Additionally, we highlight three emerging areas that we believe warrant future research attention instead.

Specifically, we have identified three areas that are commonly regarded as hot topics, but we would like to emphasize their inherent limitations:

(a) Behavioral warehouse research: Integrating the human factor is certainly one of the hot topics of warehousing research in the recent years, which is well documented by the survey papers of Grosse et al. (2017), Grosse et al. (2015), and De Lombaert et al. (2023), dedicated to this topic. We too see good reasons for this focus, because numerous empirical studies, using lab, field, and natural experiments, have shown that human factors greatly influence individual picking performance. Examples are search capabilities in front of mixed shelves (Batt & Gallino, 2019), the right incentives (De Vries et al., 2016), and fairness perception (Bai et al., 2022). Building up on these empirical results, integrating human factors into OR models is therefore also a valid contribution to warehousing research. Examples for this stream of research are provided by Matusiak et al. (2017), who integrate worker-specific picking skills into batching, Wang et al. (2022), who account for schedule-induced fluctuations in picker speeds, and Sheu and Choi (2023), who focus on schedule-induced fatigue of human pickers cooperating with robots. However, we also want to make a disclaimer. Operations researchers should not withdraw behind their models and algorithms but also think about the moral and legal limits of their research. For instance, data on the individual capabilities of human order pickers may come in handy when optimizing the division of labor in order to improve the overall picking throughput. However, neither is (for good reason) the utilization of individualized performance data legal in many parts of the world nor does it always seem fair to punish (lowly paid) pickers with higher capability by assigning them more picks with extra physical effort. Thus, enriching OR models by human factors is certainly one of the hot topics at the moment, but moral and legal aspects should certainly not be forgotten.

(b) Integration of energy aspects: The electrification of the transportation sector is seen as one elementary cornerstone to reduce mankind's greenhouse gas footprint. Thus, integrating the limited range of electric vehicles and their need to recharge into established optimization problems such as vehicle routing is certainly an important and challenging research task for operations researchers. Although electric vehicles have a much longer tradition in warehousing, where the exhaust fumes of combustion engines inside facilities are not acceptable, it is our subjective impression that we witness a spill-over effect and more and more warehousing optimization tasks also get extended by energy aspects. We agree that this can be a valid research task, but also want to add a caveat why energy aspects do not automatically have the same importance in warehousing than in road-based transportation. (i) The size of a typical warehouse is limited compared to the much longer travel distances in road networks. In relation to their operating ranges, the travel legs of in-house vehicles (e.g., forklifts, motorized picking carts, and AVGs) are short. Hence, for many (especially operational) problem settings the planning horizon is too short and charging is only required after multiple planning runs. In some settings, the charge added during closing time may even be sufficient to keep the electric vehicles operational during the whole day. (ii) The limited size of a typical warehouse also considerably reduces the recovery costs for vehicles stranded before reaching a charging device. (iii) Finally, many

vehicles applied inside warehouses are (comparatively) light-weighed and standardized, so that the invest into additional vehicles and batteries, to step in while others recharge, is lower. Hence, there is less reason for the well-known "range anxiety" in warehouses, and researchers should first think whether integrating energy aspects into a warehousing optimization problem is not only a methodological challenge but also a valid contribution from the practitioner's perspective.

(c) Holistic research: Many recent survey papers in the warehousing area (e.g. Boysen et al., 2021, 2019b; Van Gils et al., 2018) identify extended models that integrate multiple planning tasks as a fruitful field for future research. Seeing the massively improved capabilities of state-of-the-art solvers and the significant gains richer problem settings promise, we too agree that more research on holistic problems is valuable. However, we also want to remind the scientific warehousing research community that not all extensions of an existing model are actually of particular practical relevance. Even if improved solvers allow to solve extended models with sufficiently small optimality gaps, there are still organizational borders, varying scopes of responsibility, and limited data availabilities within enterprises that withstand the practical implementation of extended models. Hence, an important research contribution lies in pinpointing planning tasks that can be decomposed into independently solved models, while still maintaining substantial solution quality in comparison to a comprehensive model. Such an endeavor can pave the way for the creation of clearly structured and streamlined planning systems, thereby mitigating undue organizational complexities. Regrettably, current warehousing research tends to undervalue this particular facet.

After these (grave) words on limitations of vividly researched warehousing areas, we proceed with areas that could become the hot topics for the next years of warehousing research.

(a) Multi-purpose warehousing systems: Flexibility is an important issue in many businesses, also in warehousing. Foremost, we see demand for OR research in order to enable flexible reactions within warehouses against the following external influence. (i) Varying demand: Warehouses face hugely varying workloads, both over the week and the year (Boysen et al., 2019b). Automated parts-to-picker systems, based on fixedly installed hardware, are rather inflexible and thus need to be dimensioned such that they can handle peak loads. The costly hardware, however, then remains idle during the times of lower demand. Human-centric picker-to-parts warehouses are much more flexible and can be adapted by hiring non-permanent staff during peak times. An idea to combine the strengths of both worlds could be to apply highly automated warehouses for the base load and additional human-centric warehouses for peak times (Schiffer et al., 2022). OR models that integrate these aspects, for instance, into location planning are yet missing. Another possible reaction on varying demands is the application of shared warehouses subcontracted by retailers for their outsourced fulfillment services. If these retailers have their peak loads at different times (of the week and year), then a capacity pooling effect eases the forecast of the required warehousing capacities (see Jamili et al., 2022). How to select suitable retailers and how to organize shared warehousing during daily operations, however, has not received much attention yet. (ii) Varying order structure: Various customer groups often produce significantly different order structures. Customer households, for instance, are known to order just a few products per online order, whereas stores bundle the demand of multiple customers and thus produce larger replenishment orders (Boysen et al., 2021). An omni-channel retailer, who services online and offline customers, must thus provide a warehousing system that is able to service completely different order structures. Decision support for the choice whether this should be done from the same warehousing system, different warehousing systems unified in one warehouse, or from separate warehouses certainly deserves future research. (iii) Varying lead times: Finally, warehouses also face orders of diverging urgency. Many retailers discriminate their delivery services according to promised delivery times. This adds the decision when to phase in non-urgent orders and when to better reserve the capacity for urgent orders that may appear in the near future. This decision is especially challenging in inner-city dark stores where some customers expect within-the-hour deliveries, others on time deliveries within an agreed time window, and some orders are non-urgent. More research on order release in a dynamic environment with orders of varying urgency is certainly desirable.

(b) Flexible working-time models: Driven by the lack of workers, especially in the aging societies of many industrialized countries, many employers are on the lookout for more flexible working-time models such as those innovated by the sharing industry. Especially routine work in a warehouse (e.g., order picking directed by a pick-by-voice solution) demands not much training and can be started and ended freely during the day. Hence, crowdsourcing of warehousing labor, analogously to Uber drivers, may be possible (at least where legislation allows). This can be workers hired on short notice to extend the permanent workforce before special sales occasions such as Singles' Day. But warehousing workers can also take back deliveries on their way back home to neighboring customers (Boysen et al., 2022a), or store customers can earn extra money by picking online orders in parallel to their own purchases in a store (Dayarian & Pazour, 2022). The main new research questions these flexible working-time models add are the following: How to achieve reliable order fulfillment services if labor availability of crowdsourced workers is uncertain? How to level the workload between different departments in a warehouse on short notice, if the worker supply frequently alters unexpectedly?

(c) New warehousing systems: Finally, technical innovations produce new and adapted warehousing systems. Specifically, we see demand for future research on the following systems that already exist but have not attracted much research yet. (i) Robotic sorting is a novel system for order consolidation or parcel sortation that is based on a fleet of AMRs. Instead of relying on traditional (and rather inflexible) conveyor-based sorting systems, in a robotic sorting system, AMRs pick up individual products from a conveyor, deliver them toward a bin or scuttle in the floor currently assigned to the product's order, and hand the product over (e.g., by tilting a tray mounted on top of the AMR). These systems are already applied (and researched) for parcel sortation in distribution centers of postal service providers, where there is continuous demand for destinations and the assignment to scuttles remains stable for an extended time span (Zou et al., 2021). However, these systems can also be applied to sort picked products according to orders, where the order-to-scuttle assignment is only temporary (Boysen et al., 2023c). Since these systems have to coordinate dozens of robots and their paths along the shop floor in real-time, they constitute a challenging planning environment that demands more research. (ii) A pocket sorter is an alternative solution for order consolidation. Here, picked products are isolated into pockets hanging from an overhead conveyor (see also Section 3.2). Within their pockets, products are moved along a complex system of switches and loops, so that they can be sorted into any desired sequence. In this way, ordered products are delivered toward packing stations where the orders are prepared for shipping. However, a pocket sorter can also be applied to store fast-moving products that are anticipated to be ordered in the near future on spec. When orders arrive, only the missing products not already stored forward in the pocket sorter need to be picked. If the products to be stored forward are properly forecast, this system setup can greatly reduce order completion times. Although several large European online retailers apply this setup, research (e.g., on the dynamic selection of products stored forward) is yet missing. (iii) There are novel robotic systems that enable hybrid picker-to-parts and parts-to-picker processes. AMR (developed by, e.g., Haipick, Hikrobot, and Savoye) move autonomously through the warehouse and load multiple bins filled with SKUs from storage racks into their storage bays in a picker-to-parts process. Once a storage bay is filled, an AMR moves to a workstation where a stationary picker retrieves the ordered products from the bins in the robot's storage bay. Afterward, the AMR delivers the bins back into the rack system. Future research on all decision areas is required: from an adequate

sizing of the robots' storage bays up to a scheduling of the operational process. (iv) Another hybrid system is enabled by a crane-supplied pick face (Schwerdfeger & Boysen, 2017). The pick face in these systems is not large enough to house the complete SKU assortment, so that an automated storage and retrieval machine (e.g., a crane) loads a subset of currently demanded SKUs into a dynamic pick face in a parts-to-picker process. Along the extended pick face, a human picker travels in a picker-to-parts manner and collects the demanded products. Again, future research should address the appropriate sizing of the pick face up to the scheduling of the operational process.

To conclude, 50 years of warehousing research have generated a tremendous body of OR knowledge which is widely applied in real-world warehouses. Seeing the manifold future research opportunities, the great relevance of warehousing in e-commerce and omni-channel supply chains, and the vivid research community in this area, we would love to read the next warehousing survey paper on the fully-automated warehouses of the 4th generation written in honor of EJOR's 100th anniversary.

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